

WHAT YOU WILL BUILD THIS SEMESTER

The actual Phase-0 Autonomous MicroDrone Prototype

YOU WILL BUILD

- The real, physical drone
- The Jetson Nano onboard AI system
- The computer-vision pipeline
- The motion engine (rise / tilt / rotation + interior micro-motion)
- The safety layer
- The docking system
- The full Phase-0 deterrence prototype



SYSTEM MODULES (STUDENT TEAMS)

- **Computer Vision Team** — Species detection, approach-angle detection, interior-cart detection (Python, OpenCV, Jetson Nano)
- **Motion Engine Team** — Rise / tilt / rotation logic, interior micro-motion behaviors (Python, control algorithms)
- **Safety Layer Team** — Human detection, no-launch zones, interior mode logic (Python, rule-based systems)
- **Docking & Electronics Team** — Dock sensor logic, charging contacts, alignment system (embedded systems, GPIO, hardware integration)
- **Simulation Team** — Build the Gazebo/Webots environment for testing the motion engine (ROS2 optional, Python, simulation tools)

ARCHITECTURE OVERVIEW: Sensors → Jetson Nano → CV → Decision Layer → Motion Engine → Safety Layer → Docking System

OUTCOME: A working autonomous deterrence drone prototype built entirely by the RCOS team.

HOW YOU CAN CONTRIBUTE IMMEDIATELY

Start coding today — build the real drone this semester.

GET STARTED TODAY

- **GitHub Repository** — Link + QR code; clear README, setup instructions, and simulation environment.
- **Simulation-First Development** — Code from your laptop — no hardware required. Test in Gazebo/Webots before deploying to the drone.
- **Join the Team** — Slack/Discord QR code, weekly stand-ups, clear module ownership, and mentorship available.

GOOD FIRST ISSUES

- Add logging
- Write documentation
- Build simple CV scripts
- Create basic motion behaviors
- Improve simulation assets



THIS SEMESTER'S GOALS

- Build the CV pipeline
- Integrate the safety layer
- Implement the motion engine
- Build the docking logic



TEAM CHANNEL

All team communication will be posted on GitHub.



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